

DAY 6 – Advanced Log Analysis & Event Interpretation

Objective

The objective of this session is to build a strong foundation in **advanced log analysis and event interpretation**, focusing on how Security Operations Centers (SOC) detect, investigate, and respond to security incidents using real-world log data.

By the end of this session, learners will be able to:

- Understand and interpret key log fields used in security monitoring
- Identify critical Windows Security Event IDs
- Detect suspicious patterns and attack indicators from logs
- Perform structured security investigations
- Map analysis activities to SOC analyst roles (L1, L2, L3)
- Suggest appropriate mitigation and prevention strategies

1. Key Log Fields in Security Analysis

Logs are the backbone of cybersecurity monitoring. Each log entry contains structured information that helps analysts reconstruct events.

Important Fields:

- **Timestamp**
Indicates the exact date and time of the event.
→ Helps in building timelines and identifying attack sequences
- **Source IP Address**
Represents the origin of the activity.
→ Used to identify attackers or suspicious external/internal sources
- **Destination / User**
Specifies the targeted system or user account.
→ Helps determine the impact and sensitivity (e.g., admin accounts)
- **Event Type**
Describes the action (e.g., login success, login failure, process execution)
→ Helps classify events as normal or suspicious

- **Process / Service**

Indicates which service or application was involved (SSH, HTTP, etc.)

→ Useful for detecting service-based attacks

Importance:

Understanding these fields enables analysts to **correlate events, identify anomalies, and detect threats effectively.**

2. Windows Security Event IDs

Windows systems generate event logs that are critical for monitoring user activity and system behavior.

Event ID	Description	Security Significance
4624	Successful Login	Normal activity but must be verified if unusual
4625	Failed Login Attempt	Indicates possible brute force or unauthorized access
4672	Admin Privileges Assigned	High-risk event (privileged access)
4688	Process Creation	Useful for detecting malware or suspicious execution

Importance:

Monitoring these events helps in identifying:

- Unauthorized access attempts
- Privilege escalation
- Suspicious process execution

3. Detecting Patterns in Logs

Cyber attacks often follow patterns. Identifying these patterns is crucial in SOC operations.

Common Attack Indicators:

- **Multiple Failed Login Attempts**
→ Indicates a **Brute Force Attack**

- **Login from Unusual IP / Location**
→ Suggests **Suspicious or Unauthorized Access**
- **Successful Login After Multiple Failures**
→ Indicates **Account Compromise**
- **Unusual Process Execution**
→ Possible **Malware or Unauthorized Activity**

Key Concept:

SOC analysts focus on **behavior patterns**, not just individual events.

4. Example Log Analysis

Log Data:

Apr 10 10:17:45 Failed password for admin from 185.12.45.10

Apr 10 10:18:02 Failed password for admin from 185.12.45.10

Apr 10 10:18:15 Failed password for admin from 185.12.45.10

Apr 10 10:19:10 Accepted password for admin from 185.12.45.10

Detailed Analysis:

- Multiple failed login attempts within a short time
- Same source IP address used repeatedly
- Targeting a privileged account (**admin**)
- Final login attempt is successful

Conclusion:

This clearly indicates a **Brute Force Attack resulting in Account Compromise**

Risk Level:

High Risk

Because:

- Admin account is compromised
- Attacker now has elevated access

5. Analysis Approach

A professional SOC investigation follows a step-by-step approach:

1. **Identify anomalies**
Detect repeated failed login attempts
2. **Analyze source IP**
Check if IP is external, unknown, or suspicious
3. **Check login outcome**
Determine if login eventually succeeded
4. **Determine attack type**
Based on pattern → Brute Force Attack
5. **Assess impact and risk**
Admin account compromise → Critical
6. **Document findings**
Record observations for reporting and future reference

6. SOC Workflow Mapping

Security incidents are handled in layers within a SOC:

L1 Analyst (Monitoring Level)

- Monitors alerts and logs
- Identifies suspicious patterns
- Escalates potential incidents

L2 Analyst (Investigation Level)

- Performs detailed analysis
- Confirms attack type
- Takes action (block IP, reset password)
- Confirms brute force and blocks IP , Forces password reset

L3 Analyst (Advanced Level)

- Develops detection rules
- Improves SIEM correlation logic
- Enhances overall security posture

Assignment

1. Attack Identification

The observed activity represents a **Brute Force Attack**

2. Investigation Summary

- Multiple failed login attempts detected
- Same IP used repeatedly
- Successful login after failures
- Indicates unauthorized access

3. SOC Role Mapping

- **L1** → Detect and escalate alert
- **L2** → Investigate, block attacker, secure account
- **L3** → Create detection rules and improve monitoring

4. Prevention Techniques

- **Enable Multi-Factor Authentication (MFA)**
Adds an extra layer of security beyond passwords
- **Implement Account Lockout Policies**
Locks account after multiple failed login attempts

Conclusion

This session provided in-depth knowledge of how **log analysis and event interpretation** are performed in real-world SOC environments.

By understanding:

- Log structures
- Event patterns
- Attack indicators

learners can effectively detect threats such as brute force attacks and respond appropriately.

Strong log analysis skills are essential for roles such as:

- SOC Analyst
- Security Analyst
- Incident Responder

This knowledge forms a critical foundation for advanced topics like:

- Threat Hunting
- Incident Response
- Digital Forensics